

Passwords are analogous to the key in a cryptosystem that allows encryption and decryption of anything the password has access to. The principal element of this system is the password selection process. And that's a whole other subject that we can't cover here. But in a nutshell, the longer the password, the more random it will be and the harder it is to guess. So if you think it's easy for you to remember, you should know that it will be all the easier to crack.

Credentialing systems

A credential is a proof of qualification or competence that is attached to a person to indicate suitability for something. Suppose you go to a bank for a loan, they check your credentials before approving the loan. Your credentials are checked not only from the paperwork, but also from your past record and your references. Your driver's license and passport are forms of credentials. Progress in the field of implementing electronic credentials has been rather slow. Electronic credentials allow electronic verification of the credence of a claim. It's not a standalone system, but is being used in conjunction with other devices such as smart cards which perform cryptographic functions and store secret information.

CIBIL (Credit Information Bureau (India) Limited) is India's first credit information bureau. It shares credit information with banks, financial institutions and credit card companies and generates Credit Information Reports.

Digital signatures

A digital signature is a mechanism by which a message is authenticated i.e. proving that a message is coming from a given sender, much like a signature on a paper document. To be as effective as a signature on paper, digital signatures must be hard to forge and accepted in a court of law as binding upon all parties to the transaction. The need for digital signatures arises when the parties dealing in a transaction are not physically close, and the volume of paperwork is high, in other words big business dealings.

Digital signatures can be created using a public key cryptosystem and hashing process. Hashing produces a message digest that is a small and unique representation of the original message. Hashing is a one-way algorithm, i.e. the message can't be derived from the digest. Let's say that Alex is sending a message to Neil. Alex first hashes the message to produce a digest, and then encrypts the digest with her private-key to create her personal signature; the public-key and hash algorithm are appended to it.